

# The Box in Balance

## Student sheet

Name: \_\_\_\_\_

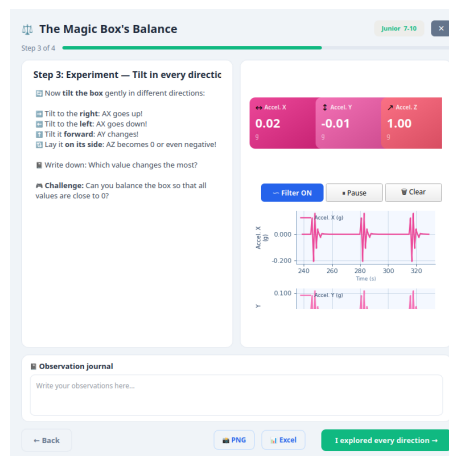
Date: \_\_\_\_\_

**Safety:** I hold the box in both hands, over the table. The cable stays slack. I never throw the box to a classmate: I put it down on the table.

### 1. What I see on the screen

On the screen you can see three numbers: **AX**, **AY** and **AZ**. The box feels whatever holds it up **pushing it upwards** so that it does not fall — the table, or your hand. The numbers show how big that push is in each of the three directions.

The activity starts with the **Start** button. You move on with the green button at the bottom right, you go back with **Back**, and at the last step the green button reads **Finish**.



### 2. What I do on my own

**First I predict. I do not change my prediction after I measure.**

1. The box is lying still on the table. I think AZ will show:

☐ zero      ☐ something other than zero      if you can, write the number too: \_\_\_\_\_

because \_\_\_\_\_

2. When I tilt the box to the right, I think AX will:

☐ go up      ☐ go down      ☐ stay the same

because \_\_\_\_\_

**Now I measure.** I put the box down, wait for it to be still, count to three, then write the numbers exactly as they appear on the screen.

| Position of the box | AX | AY | AZ |
|---------------------|----|----|----|
| Flat on the table   |    |    |    |
| Tilted to the right |    |    |    |
| Tilted to the left  |    |    |    |
| Tipped forward      |    |    |    |
| On its edge         |    |    |    |
| Upside down         |    |    |    |
| Flat, a second time |    |    |    |

When I tilt the box, the number AX: ☐ ↑ ☐ ↓ ☐ →  
(tick one: goes up, goes down, stays the same)

**Challenge.** Can you make all three numbers zero at the same time? ☐ YES ☐ NO

### 3. The conclusion I write

I predicted that \_\_\_\_\_  
\_\_\_\_\_

The data show that \_\_\_\_\_  
\_\_\_\_\_

The difference is explained by \_\_\_\_\_  
\_\_\_\_\_

**Write one sentence:** why the box does not show zero when it is sitting still.

\_\_\_\_\_

#### Exit ticket

Write one thing that surprised you today.

\_\_\_\_\_